

# EETX433A

## Crystal-less Multifunctional RF Transmitter

Low-power RF solution for wireless remote control

TRANSMIT POWER

**13 dBm**

MODULATION

**OOK / ASK**

SLEEP CURRENT

**800 nA**

TRANSMIT CURRENT

**12 mA**

VERSION

**v1.0**

PACKAGE INFORMATION

**SOP8**

DEVELOPED BY

**Engeletron Engenharia**

REVISION

**v1.3 · 2026-06-03**

## • Features

|                  |                 |
|------------------|-----------------|
| Transmit power   | 13 dBm          |
| Modulation       | OOK / ASK       |
| Sleep current    | 800 nA          |
| Transmit current | 12 mA           |
| Frequencies      | 433.92 MHz      |
| Supply voltage   | 2.0 V – 3.6 V   |
| Package          | SOP-8           |
| Certifications   | FCC, CE, ANATEL |

## • Applications

- Automatic gates and doors
- Access control
- Property and home alarms
- Automotive alarms
- Electric fences
- Fire detection
- Wireless doorbells
- Panic button
- Home and building automation
- Low-power wireless sensors (IoT)

## • Overview

The EETX433A integrates a 433 MHz PLL oscillator, an RF transmitter, and a microcontroller (8-bit MCU core) on a single chip, delivering a simpler design, more flexible encoding, lower cost, and a smaller footprint — fully compliant with FCC, CE and ANATEL standards.

A built-in key-detection auto-shutdown feature minimizes standby current. The device also integrates output overcurrent protection, on-chip over-temperature protection, and supply undervoltage protection.

Building a complete wireless transmission system with the EETX433A requires only 2 inductors, 4 capacitors, resistors, a battery, and buttons — a fully integrated solution for remote control applications.

## • Functional Block Diagram

The EETX433A integrates on a single die all blocks required for a standalone 433.92 MHz RF transmitter: reference oscillator, PLL for carrier synthesis, OOK/ASK modulator, power amplifier, 8-bit MCU with code ROM, power-management (UVLO/OTP/OCF), and operating-mode selection logic (CFG).

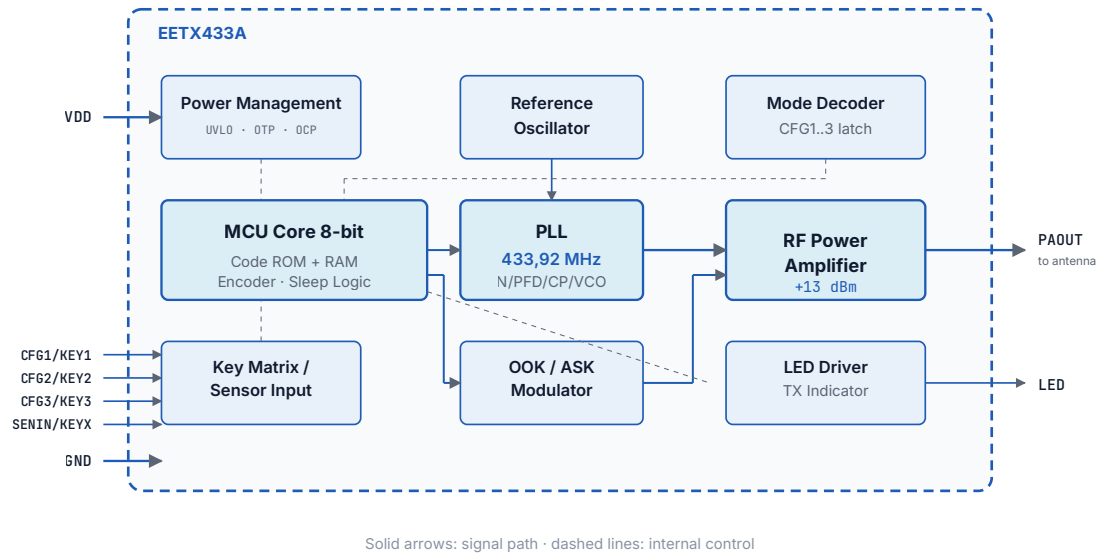


Figure 1: EETX433A internal functional block diagram

• Pinout & Pin Configuration



SOP-8 (top view)

• Pin Description

| Pin | Name       | I/O | Function                                                                                                                                                                            |
|-----|------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | GND        | P   | Power ground reference.                                                                                                                                                             |
| 2   | PAOUT      | O   | Power amplifier (PA) output.                                                                                                                                                        |
| 3   | LED        | O   | Data transmission indicator (LED driver).                                                                                                                                           |
| 4   | CFG3/KEY3  | I   | Input pin. At power-up it sets operating-mode/sensor-type Configuration 3 (CFG3). In Mode 0 it is the direct key input KEY3 and also participates in the key matrix (see Figure 5). |
| 5   | CFG2/KEY2  | I   | Input pin. At power-up it sets operating-mode/sensor-type Configuration 2 (CFG2). In Mode 0 it is the direct key input KEY2 and also participates in the key matrix (see Figure 5). |
| 6   | CFG1/KEY1  | I   | Input pin. At power-up it sets operating-mode/sensor-type Configuration 1 (CFG1). In Mode 0 it is the direct key input KEY1 and also participates in the key matrix (see Figure 5). |
| 7   | SENIN/KEYX | I/O | Input/output pin. Modes 1–7: digital sensor input (SENIN). Mode 0: acts only as KEYx — a digital output (level 0/1) that drives the key matrix for keys 4, 5 and 6.                 |
| 8   | VDD        | P   | Positive supply voltage.                                                                                                                                                            |

• **Configuration Modes (CFG Pins)**

Beyond operating as a transmitter with up to 6 keys, the EETX433A can be configured as a magnetic opening sensor (door/window) or a passive PIR motion sensor. Configuration is set by tying pins CFG1, CFG2 and CFG3 to GND (level 0) or VDD (level 1) per the table below:

Pins CFG1, CFG2 and CFG3 are sampled in hardware ONLY at power-up. The binary combination of the three pins (weight CFG3:CFG2:CFG1) selects one of 8 operating modes. Dynamic changes during operation are ignored. The SENIN pin becomes the external sensor input in modes 1–7. See "Functional Operation" for details.

- **Transmitter Mode (Mode 0)** — All three CFG pins are left open during power-up (no connection to VDD/GND) — this selects Mode 0. The IC works as a keypad of up to 6 keys: up to 3 keys wired directly (one per CFG pin) and keys 4, 5 and 6 through control of the KEYx pin (see "Key Mapping" and the 4- and 6-key schematics, Figures 4 and 5). Each key transmits a unique 4-bit value.
- **Magnetic Mode (Modes 1, 2, 3, 6)** — The EETX433A monitors the SENIN pin connected to a magnetic sensor (e.g. TMR, Reed switch). Ideal for door/window opening detection.
- **Passive / PIR Mode (Modes 4, 5, 7)** — The EETX433A monitors the SENIN pin connected to a pyroelectric presence sensor (PIR). Must be a DIGITAL pyrosensor with ON/OFF (high/low) output. Ideal for motion detection.

| MODE | TYPE        | TRANSMITS | CFG3   | CFG2   | CFG1   | SENIN   |
|------|-------------|-----------|--------|--------|--------|---------|
| 0    | TRANSMISSOR | TECLA     | Aberto | Aberto | Aberto | Aberto  |
| 1    | MAGNÉTICO   | V1B3      | 0      | 0      | 1      | Entrada |
| 2    | MAGNÉTICO   | V3B1      | 0      | 1      | 0      | Entrada |
| 3    | MAGNÉTICO   | A4F5B3    | 0      | 1      | 1      | Entrada |
| 4    | PASSIVO     | V1B3      | 1      | 0      | 0      | Entrada |
| 5    | PASSIVO     | V3B1      | 1      | 0      | 1      | Entrada |
| 6    | MAGNÉTICO   | V2B1      | 1      | 1      | 0      | Entrada |
| 7    | PASSIVO     | V2B1      | 1      | 1      | 1      | Entrada |

Legend: V = Violation · B = Low battery · A = Open · F = Close. Notation: the number after each letter is the 4-bit value (D3D2D1D0) transmitted in the data field when that event occurs. Example — Mode 1 (V1B3): on a Violation the transmitter sends data = 0001 (= 1); when the sensor reports Low Battery, data = 0011 (= 3) is sent. The receiver decodes the 4-bit field to identify the event. Violation (V) and Open/Close (A/F) are distinct operating modes — a given unit never sends both. In Open/Close modes, Open corresponds to a 0→1 transition of SENIN and Close to a 1→0 transition. Since the frame carries no mode number, interpretation of the 4 bits depends on the unit's configured mode (the same value means different events in different modes); the unique 20-bit address identifies the unit at the receiver.

## • Block Diagram & Typical Application

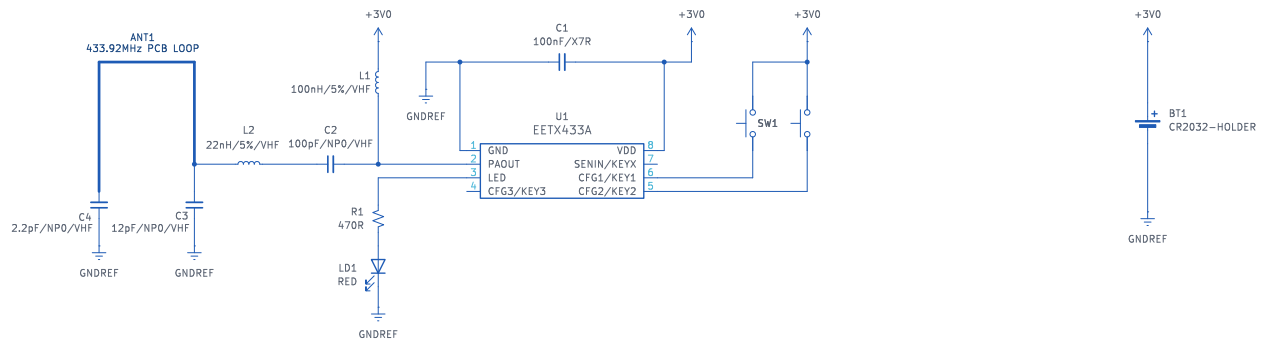


Figure 2: 433.92 MHz typical application — 2 buttons

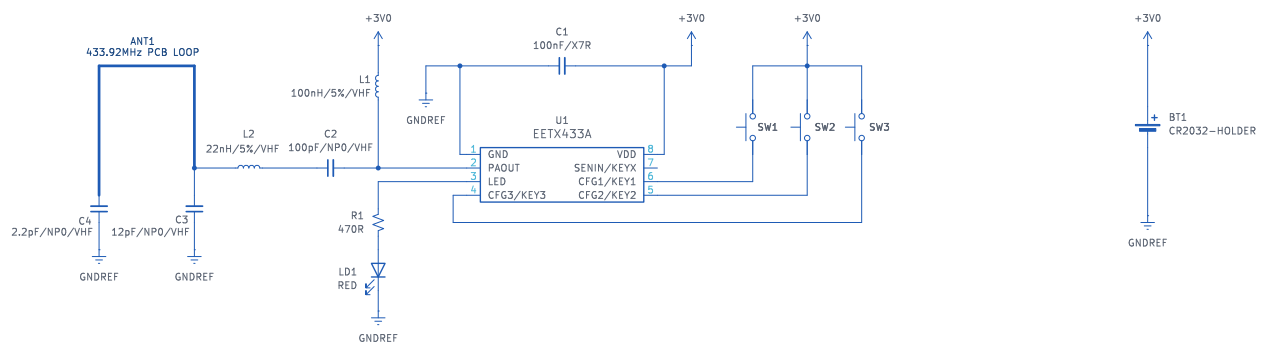


Figure 3: 433.92 MHz typical application — 3 buttons

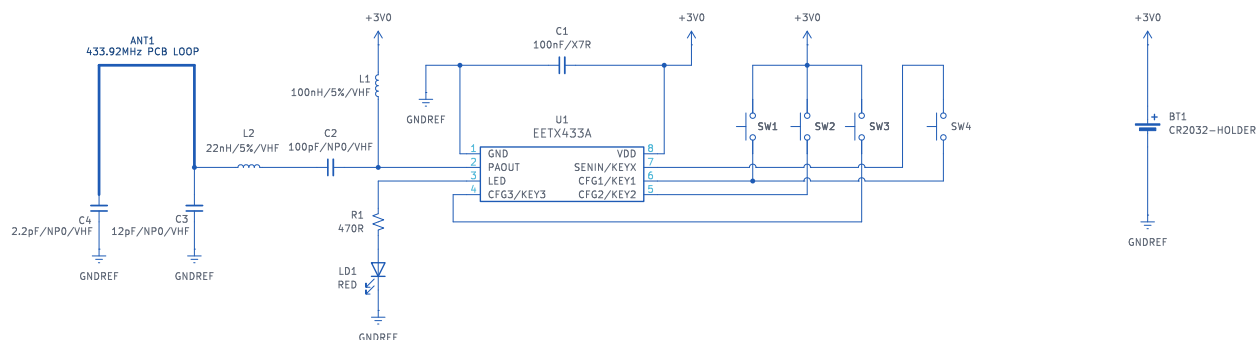


Figure 4: 433.92 MHz typical application — 4 buttons

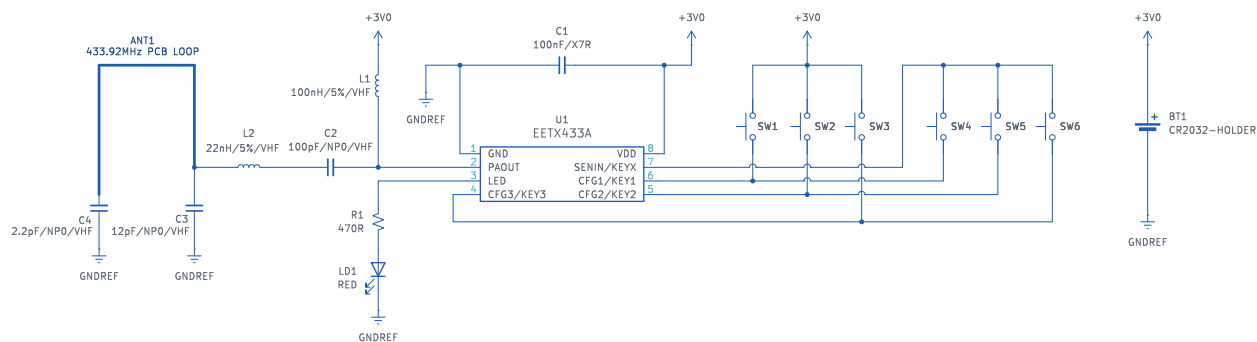


Figure 5: 433.92 MHz typical application — 6 buttons

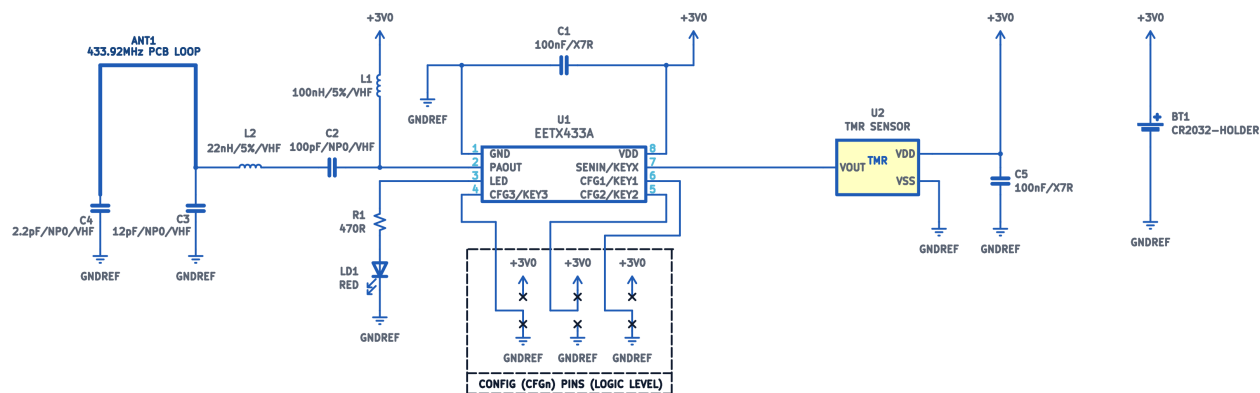


Figure 6: Typical application as magnetic opening sensor (TMR + EETX433A)

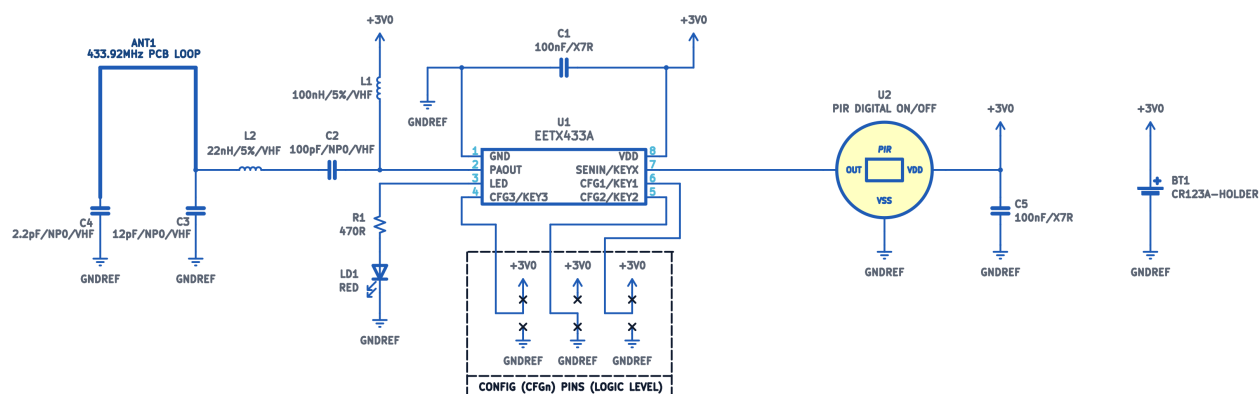


Figure 7: Typical application as motion sensor (digital ON/OFF PIR + EETX433A)



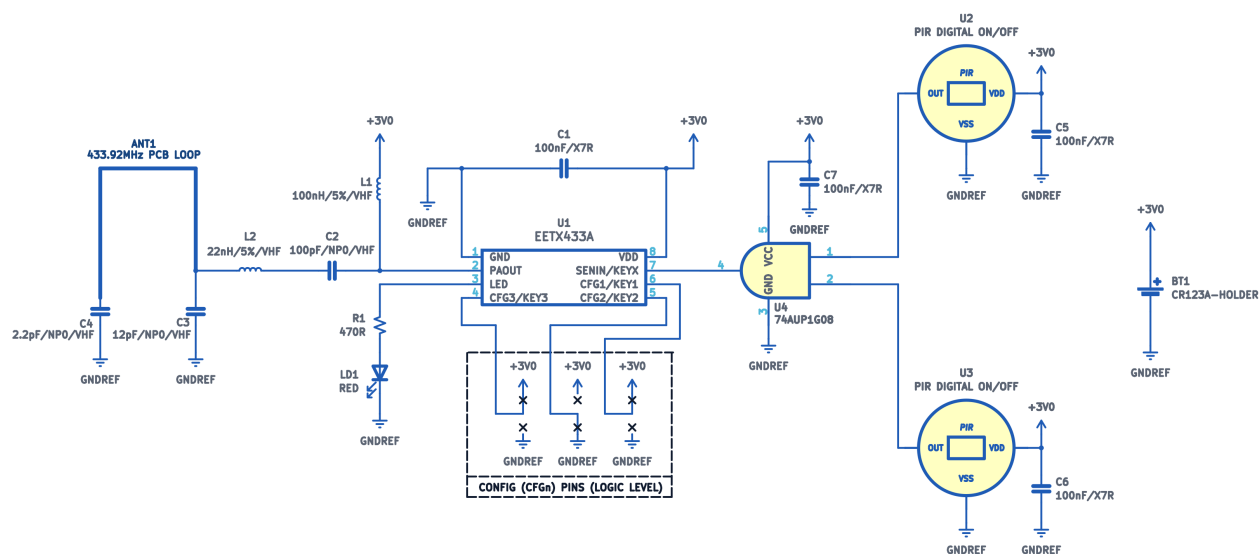


Figure 8: Typical application as dual-lens motion sensor (2× digital PIR + ultra-low-power AND logic gate + EETX433A)

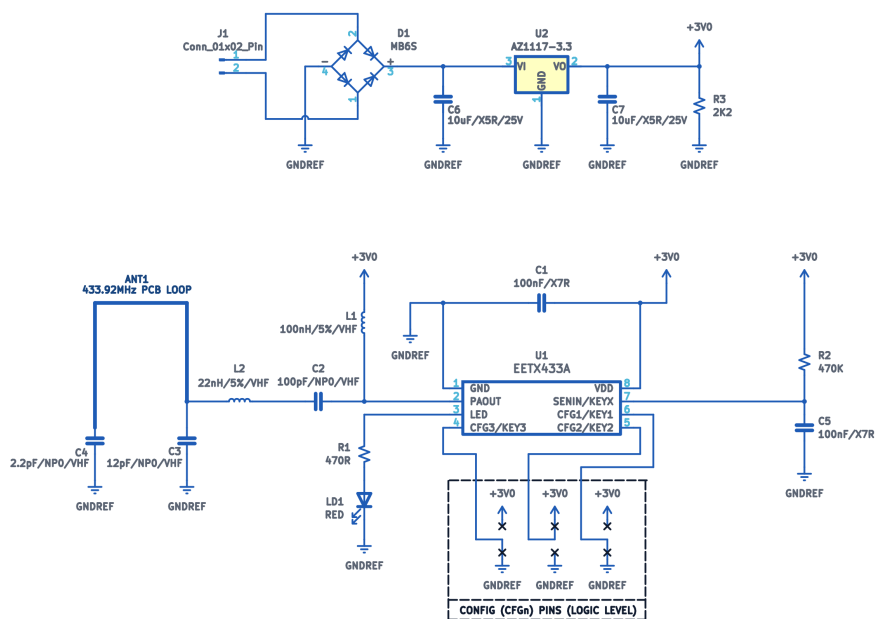


Figure 9: Typical application — transmitter installed on the high-beam headlight (bridge rectifier + 3.3 V LDO + EETX433A)

## • Data Format

The EETX433A transmits using a pulse-width OOK encoding scheme at 433.92 MHz. Each code word contains a pilot, a start, address, data and an anti-code. The basic time unit ( $\lambda$ ) equals 1/3 of the bit period; in the EETX433A,  $\lambda = 500 \mu\text{s}$ , so each bit lasts 1.5 ms. In OOK modulation, a high level = carrier present (carrier-on) and a low level = carrier absent (carrier-off).

In Transmitter mode, transmission triggers when a key connects the KEY pin to VDD (logic 1). In sensor modes (1–7), transmission triggers on the rising edge (0→1) of the SENIN pin. See "Functional Operation" for details.

### Packet Structure

Each packet ( $\approx 54 \text{ ms} \cdot 108\lambda$ ) consists of:  $23\lambda$  pilot (level 0) +  $1\lambda$  start (level 1) + 20-bit programmable address ( $60\lambda$ ) + 4-bit data (D3 D2 D1 D0,  $12\lambda$ ) + 4-bit fixed anti-code 0101 ( $12\lambda$ ). The pilot ( $23\lambda$  at level 0, silent) followed by the start ( $1\lambda$  at level 1) synchronises the start of each repeated packet. Each field is transmitted starting from the first bit shown: the address from A0 to A19 and the data from D3 to D0. The 0101 anti-code is fixed and used for validation — the receiver discards the packet if it does not match.

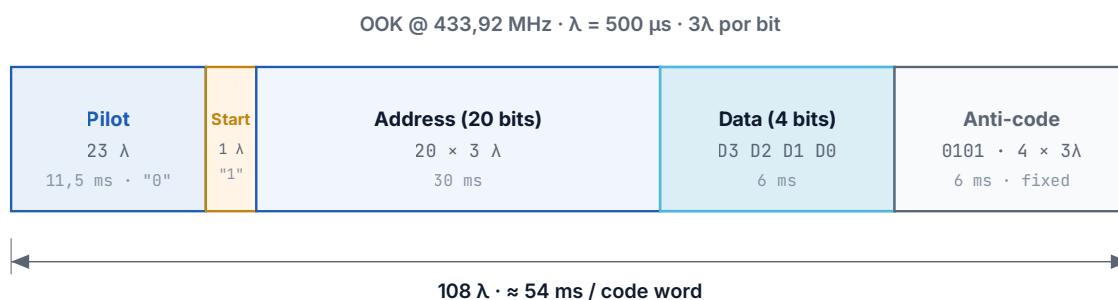


Figure 10: Transmission packet structure

## Complete Frame Timing

The diagram below shows the internal fosc clock and the DOUT data line aligned across the phases of the code word. Each address/data/anti-code bit occupies  $3\lambda$ ; the pilot occupies  $23\lambda$  (level 0), the start  $1\lambda$  (level 1) and the anti-code  $12\lambda$  (4 fixed bits 0101).

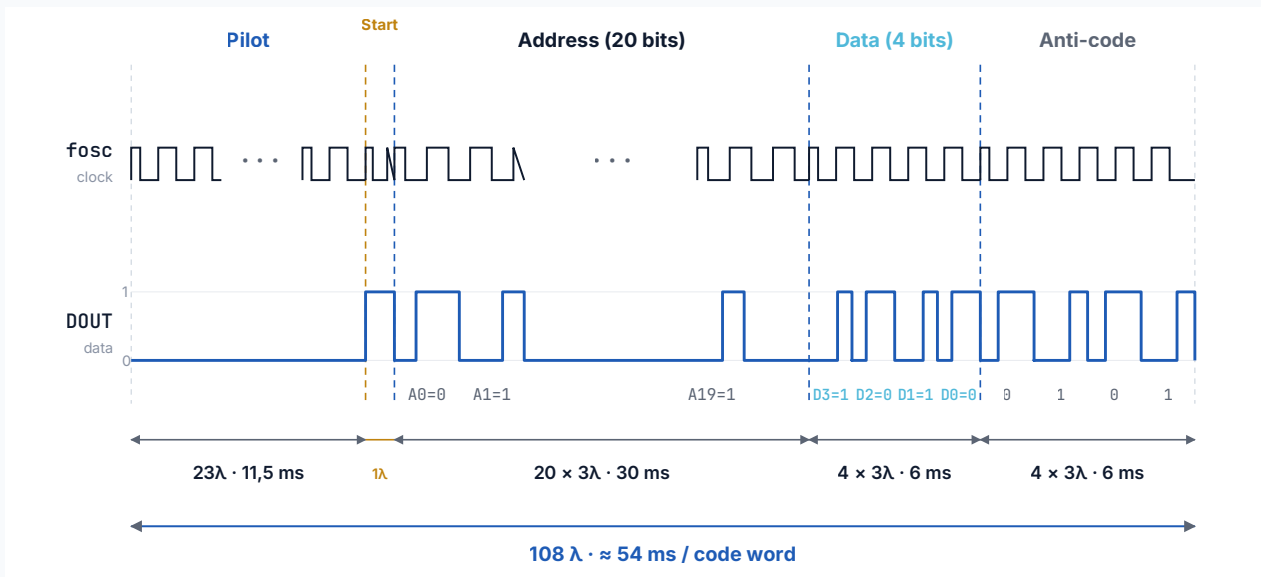


Figure 11: Frame timing — fosc + DOUT (Pilot · Start · Address · Data · Anti-code)

## Logic Bit — Waveform

Each address/data bit occupies  $3\lambda$  (1.5 ms). The logic level is set by the ratio between the low and high times:

- Bit "0":  $1\lambda$  low +  $2\lambda$  high (500  $\mu\text{s}$  low / 1000  $\mu\text{s}$  high)
- Bit "1":  $2\lambda$  low +  $1\lambda$  high (1000  $\mu\text{s}$  low / 500  $\mu\text{s}$  high)

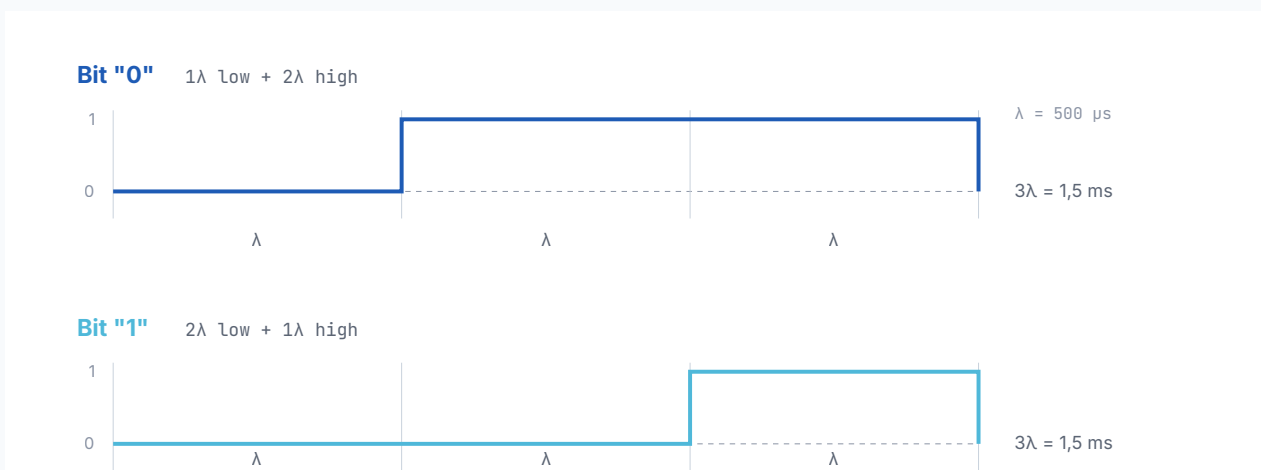


Figure 12: Logic-bit waveforms (OOK modulation)

Characteristic Times

| Parameter                                  | Símbolo   | Typ.         | Unit          |
|--------------------------------------------|-----------|--------------|---------------|
| Basic unit ( $\lambda = 1/3$ of bit clock) | $\lambda$ | 500          | $\mu\text{s}$ |
| Bit duration ( $3\lambda$ )                | t_BIT     | 1,5          | ms            |
| Pilot ( $23\lambda$ , level 0)             | t_PIL     | 11,5         | ms            |
| Start ( $1\lambda$ , level 1)              | t_ST      | 0,5          | ms            |
| Address (20 bits)                          | t_ADDR    | 30           | ms            |
| Data (4 bits)                              | t_DATA    | 6            | ms            |
| Anti-code (4 bits, $12\lambda$ )           | t_ANTI    | 6            | ms            |
| Complete packet (typ.)                     | t_PKT     | $\approx 54$ | ms            |

## • Functional Operation

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This section describes the operational behavior of the EETX433A for designers using the part. All parameters listed below are fixed in the embedded MCU firmware and cannot be altered in production.

### Factory-Programmed Address

The 20-bit address is programmed at the factory during EETX433A production and is unique per order code. No additional programming step is required on the customer PCB — the IC is delivered ready-to-use, in a 4 000-piece reel sharing the same address. To order batches with different addresses, contact Engelertron.

### Transmission Trigger

Mode 0 (Transmitter): transmission triggers on a key press. Up to 3 keys connect a CFG pin (CFG1, CFG2 or CFG3) directly to VDD (logic 1); keys 4, 5 and 6 use control of the KEYx pin (SENIN/KEYx) — see "Key Mapping" and Figures 4 and 5. Transmission continues while the key is held, with a minimum duration of 864 ms (16 packets) and a maximum of 5400 ms (100 packets). If the user releases the key before 864 ms, the IC still completes the minimum duration.

Modes 1–7 (Sensor): transmission triggers on the rising edge (0 → 1 transition) of the SENIN pin, coming from a digital sensor with ON/OFF output (magnetic TMR or digital PIR sensor).

Transmission duration is fixed at 1 296 ms (24 packets) regardless of the sensor state after the trigger.

### PIR Mode Re-arming (Motion)

In PIR (Passive) modes, after a transmission the IC enters a lockout state until the sensor remains at logic low (no motion detected) for at least 120 seconds (2 minutes) continuously. This interval prevents transmission bursts in environments with continuous motion (e.g. wind-blown curtains). Only after this idle period will the IC accept a new SENIN rising edge and trigger a new transmission.

### Low-Battery Detection (sensor modes 1–7)

In sensor modes (1–7), the EETX433A continuously monitors the supply voltage. When VDD falls below  $\approx 2.2$  V (falling threshold 2.20 V; re-arm when VDD rises above  $\approx 2.23$  V — hysteresis  $\approx 30$  mV; tolerance  $\pm 0.1$  V), the device starts signaling the Low-Battery event, transmitted in the 4-bit data field as the "B" code in the modes table "TRANSMITS" column. This lets the receiver alert the user to replace the battery. Low-battery detection does not apply to Mode 0 (Transmitter).

**CFG Pin Sampling**

CFG1, CFG2 and CFG3 pins are sampled ONLY at power-up (VDD ramp from 0 V to the operating level). Dynamic changes during operation are ignored. After the latch ( $\leq 1$  ms), the same pins become KEY1/KEY2/KEY3 (mode 0) or are reserved for internal use (modes 1–7). Important: in Mode 0, do not power up the device with a key pressed — that would tie a CFG pin to VDD during power-up and select an unintended sensor mode. Make sure all keys are released when power is applied (e.g., when inserting the battery). During power-up, do not apply any level to CFG1/CFG2/CFG3 or to KEYx (SENIN/KEYx) — leaving them undriven is what guarantees entry into Mode 0.

**Key Mapping (Mode 0)**

In Mode 0 (Transmitter), CFG1, CFG2 and CFG3 must be left floating (no connection to VDD or GND) at power-up — this is what selects Mode 0. Only 3 keys can be wired directly, one per CFG pin, tying the pin to VDD when pressed: KEY1 → CFG1/KEY1 (pin 6), KEY2 → CFG2/KEY2 (pin 5) and KEY3 → CFG3/KEY3 (pin 4). Keys 4, 5 and 6 are not direct: they rely on control of the KEYx pin (SENIN/KEYx), per the 4- and 6-key application schematics (see Figures 4 and 5). Each key transmits a fixed 4-bit value per the table below:

| Key  | D3 | D2 | D1 | D0 | Decimal value |
|------|----|----|----|----|---------------|
| KEY1 | 0  | 0  | 0  | 1  | 1             |
| KEY2 | 0  | 0  | 1  | 0  | 2             |
| KEY3 | 0  | 0  | 1  | 1  | 3             |
| KEY4 | 0  | 1  | 0  | 0  | 4             |
| KEY5 | 0  | 1  | 0  | 1  | 5             |
| KEY6 | 0  | 1  | 1  | 0  | 6             |

**SENIN Pin (Sensor Input)**

In modes 1–7, the SENIN pin operates as a pure digital input (high impedance). There is no internal pull-up or pull-down: the connected sensor is responsible for maintaining the defined logic level (high or low) under all operating conditions. Sensors with push-pull outputs (e.g. digital TMR, ON/OFF digital PIR) meet this requirement without additional components. For open-drain output sensors or dry contacts, use an external pull-down resistor (typical 100 k $\Omega$  → GND) to guarantee a defined low level at rest.

## Operational Timing Parameters

| Parameter                              | Symbol    | Min. | Typ. | Max. | Unit |
|----------------------------------------|-----------|------|------|------|------|
| Tempo de wake-up (power-up até 1º bit) | t_WK      | —    | 50   | 100  | ms   |
| Tempo de estabilização do PLL          | t_PLL     | —    | 0,2  | 1    | ms   |
| Duração de 1 pacote (108λ)             | t_PKT     | —    | 54   | —    | ms   |
| Modo 0 — TX mínimo (16 pacotes)        | t_TX0_min | —    | 864  | —    | ms   |
| Modo 0 — TX máximo (100 pacotes)       | t_TX0_max | —    | —    | 5400 | ms   |
| Modos 1–7 — TX fixo (24 pacotes)       | t_TX1_7   | —    | 1296 | —    | ms   |
| Rearme PIR — sem movimento contínuo    | t_REARM   | 120  | —    | —    | s    |
| Latch dos pinos CFG (power-up)         | t_CFG     | —    | —    | 1    | ms   |

## Compatible Receiver

To receive and decode the signal transmitted by the EETX433A, an ASK (Amplitude Shift Keying) RF receiver with a reception bandwidth ( $\Delta f_{RX}$ ) of  $\pm 300$  kHz or higher is recommended — comfortably covering the  $\pm 250$  kHz unit-to-unit  $f_o$  variation of the EETX433A. Recommended sensitivity is -110 dBm to -112 dBm for typical residential/commercial range. Frame decoding (23λ pilot + 1λ start + 20-bit address + 4-bit data + 4-bit anti-code 0101,  $\lambda = 500$  μs) is typically performed by a microcontroller on the receiver side.

## • Electrical Characteristics

### Absolute Maximum Ratings

| Parameter | Symbol           | Min. | Max. | Unit |
|-----------|------------------|------|------|------|
| VDD       | V <sub>DD</sub>  | -0.3 | 3.6  | V    |
| TJ        | T <sub>J</sub>   | -40  | 150  | °C   |
| TSTG      | T <sub>STG</sub> | -50  | 150  | °C   |
| ESD (HBM) | V <sub>ESD</sub> | —    | ±5   | kV   |

### Recommended Operating Conditions

| Parameter | Symbol          | Condition | Min. | Typ. | Max. | Unit |
|-----------|-----------------|-----------|------|------|------|------|
| VDD       | V <sub>DD</sub> |           | 2.0  | 3.3  | 3.6  | V    |
| Ta        | T <sub>a</sub>  |           | -40  | 25   | 85   | °C   |
| RL        | R <sub>L</sub>  |           |      | 50   |      | Ω    |

### DC Characteristics

V<sub>SS</sub>=0V, V<sub>DD</sub>=3V, T<sub>a</sub>=25°C

| Parameter | Symbol          | Condition      | Min. | Typ. | Max. | Unit |
|-----------|-----------------|----------------|------|------|------|------|
| VDD       | V <sub>DD</sub> |                | 2.0  | 3.3  | 3.6  | V    |
| IDD       | I <sub>DD</sub> | 13dBm          | 8    | 12   | 14   | mA   |
| ISD       | I <sub>SD</sub> | no key pressed | 700  | 800  | 900  | nA   |

### Analog / RF Characteristics

| Parameter         | Symbol             | Condition                     | Min. | Typ. | Max. | Unit |
|-------------------|--------------------|-------------------------------|------|------|------|------|
| Po (433M)         | P <sub>0</sub>     | RL=50Ω, VDD=3V                |      | 13   |      | dBm  |
| Po (433M)         | P <sub>0</sub>     | RL=50Ω, VDD=3.6V              |      | 14   |      | dBm  |
| Δf entre unidades | Δf <sub>unit</sub> | VDD=3V, Ta=25°C, unit-to-unit |      | ±200 | ±250 | kHz  |



## • Soldering Guidelines

The EETX433A is MSL-3 rated (J-STD-020) and withstands standard lead-free reflow soldering. Even so, do not exceed the recommended profile: excessive heat can cause thermal-deformation damage. Use a non-corrosive rosin-type flux and observe the moisture-sensitivity level (MSL-3) before soldering.

### 1. Hand soldering (test samples)

- Use a 30 W soldering iron with tip temperature 260–300 °C.
- Maximum contact time: 5 seconds.
- Keep the tip thoroughly clean.

### 2. Reflow soldering (SMT)

Follow a lead-free reflow profile per IPC/JEDEC J-STD-020: preheat/soak 150 °C to 200 °C for 60–120 s, ramp-up  $\leq 3$  °C/s, time above 217 °C (liquidus) 60–150 s, peak temperature 260 °C (max) for no more than 20–30 s, and ramp-down  $\leq 6$  °C/s. Maximum of 2 reflow cycles.

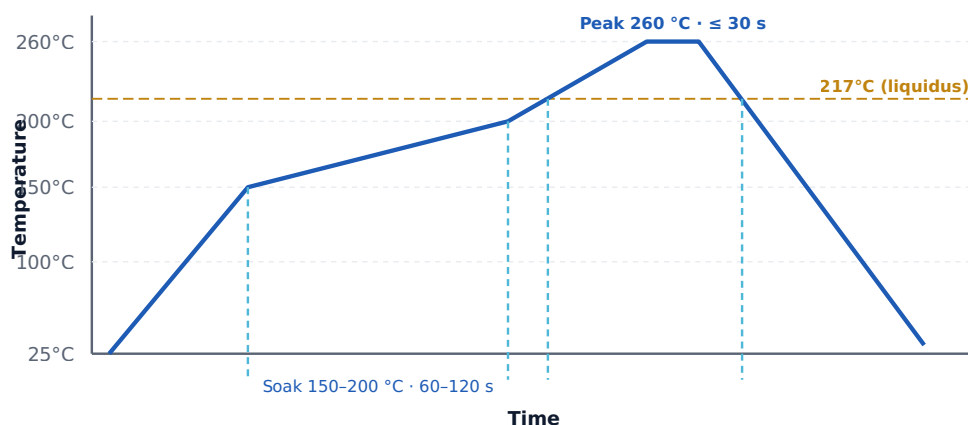


Figure 13: Recommended reflow temperature profile

• Package Information

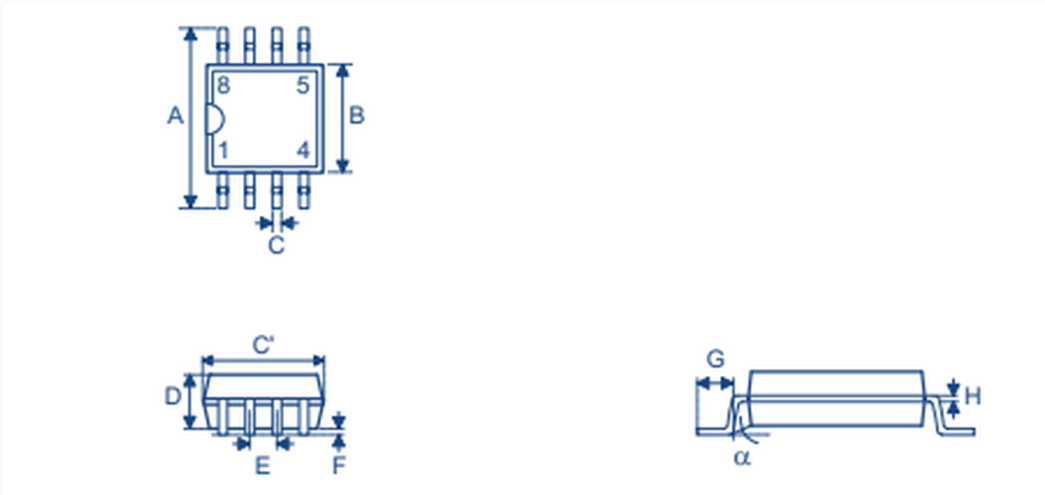


Figure 14: SOP-8 package — mechanical drawing

| Symbol   | Min (mil) | Nom (mil) | Max (mil) |
|----------|-----------|-----------|-----------|
| A        | 228       | —         | 244       |
| B        | 149       | —         | 157       |
| C        | 14        | —         | 20        |
| C'       | 189       | —         | 197       |
| D        | 53        | —         | 69        |
| E        | —         | 50        | —         |
| F        | 4         | —         | 10        |
| G        | 22        | —         | 28        |
| H        | 4         | —         | 12        |
| $\alpha$ | 0°        | —         | 10°       |

• **Packaging (Tape & Reel)**

The EETX433A is supplied in carrier tape wound on a standard 330 mm reel, fully compatible with SMT pick-and-place systems.

STANDARD QUANTITY

4 000 pieces per reel

Collective packaging: 1 reel + identification card + sealed anti-static bag, per MSL-3 / J-STD-020.

**Reel Dimensions — SOP-8**

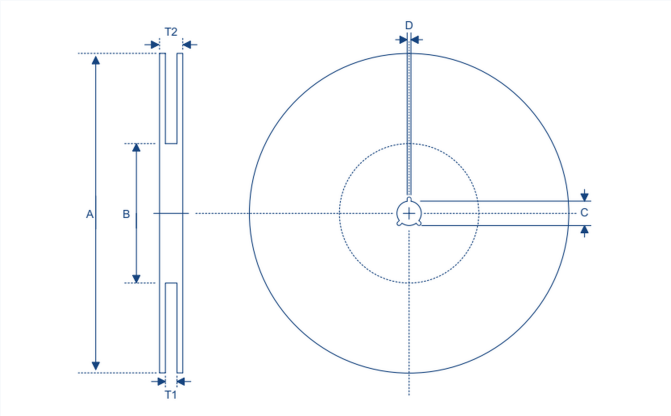


Figure 15: Standard 330 mm reel — 4 000 pcs

| Symbol | Description                                      | Dimension (mm)   |
|--------|--------------------------------------------------|------------------|
| A      | Diâmetro externo do rolo / Reel outer diameter   | 330 ±1,0         |
| B      | Diâmetro interno do rolo / Reel inner diameter   | 62 ±1,5          |
| C      | Diâmetro do furo central / Spindle hole diameter | 13,0 +0,5 / -0,2 |
| D      | Largura do entalhe-chave / Key slit width        | 2,0 ±0,15        |
| T1     | Espaço entre flanges / Space between flanges     | 12,8 +0,3 / -0,2 |
| T2     | Espessura total do rolo / Reel thickness         | 18,2 ±0,2        |

Carrier Tape Dimensions — SOP-8

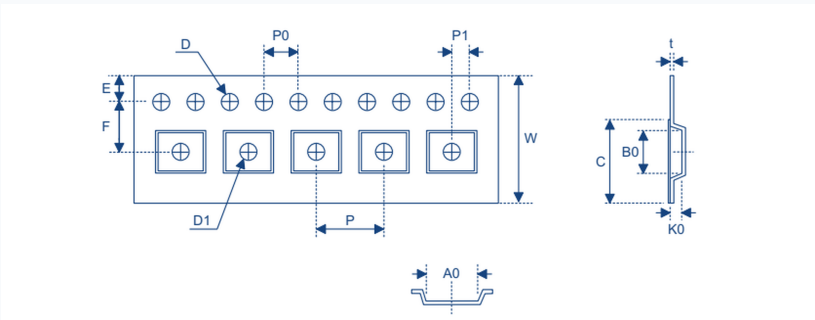


Figure 16: 12 mm carrier tape

| Symbol | Description                                         | Dimension (mm)   |
|--------|-----------------------------------------------------|------------------|
| W      | Largura da fita / Carrier tape width                | 12,0 +0,3 / -0,1 |
| P      | Passo dos compartimentos / Cavity pitch             | 8,0 ±0,1         |
| E      | Posição da perfuração / Perforation position        | 1,75 ±0,1        |
| F      | Cavidade → perfuração (largura) / Cavity to perf    | 5,5 ±0,1         |
| D      | Diâmetro da perfuração / Perforation diameter       | 1,55 ±0,1        |
| D1     | Diâmetro do furo da cavidade / Cavity hole diameter | 1,5 +0,25        |
| P0     | Passo da perfuração / Perforation pitch             | 4,0 ±0,1         |
| P1     | Cavidade → perfuração (comprimento)                 | 2,0 ±0,1         |
| A0     | Comprimento da cavidade / Cavity length             | 6,4 ±0,1         |
| B0     | Largura da cavidade / Cavity width                  | 5,20 ±0,1        |
| K0     | Profundidade da cavidade / Cavity depth             | 2,1 ±0,1         |
| t      | Espessura da fita / Carrier tape thickness          | 0,3 ±0,05        |
| C      | Largura da fita de cobertura / Cover tape width     | 9,3              |

For more information about this product, visit: <https://engeletron.com.br/>

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